midterm_dsII

2026-03-25

```r
library(tidymodels)
library(caret)
library(pls)
library(earth)
library(mgcv)
library(splines)
library(corrplot)
library(car)
library(patchwork)
```

1. Load, clean and EDA analysis

Load, clean, and prepare dataset A for creating potential predictive models.

```r
load("data/datA.RData")
summary(datA)
```

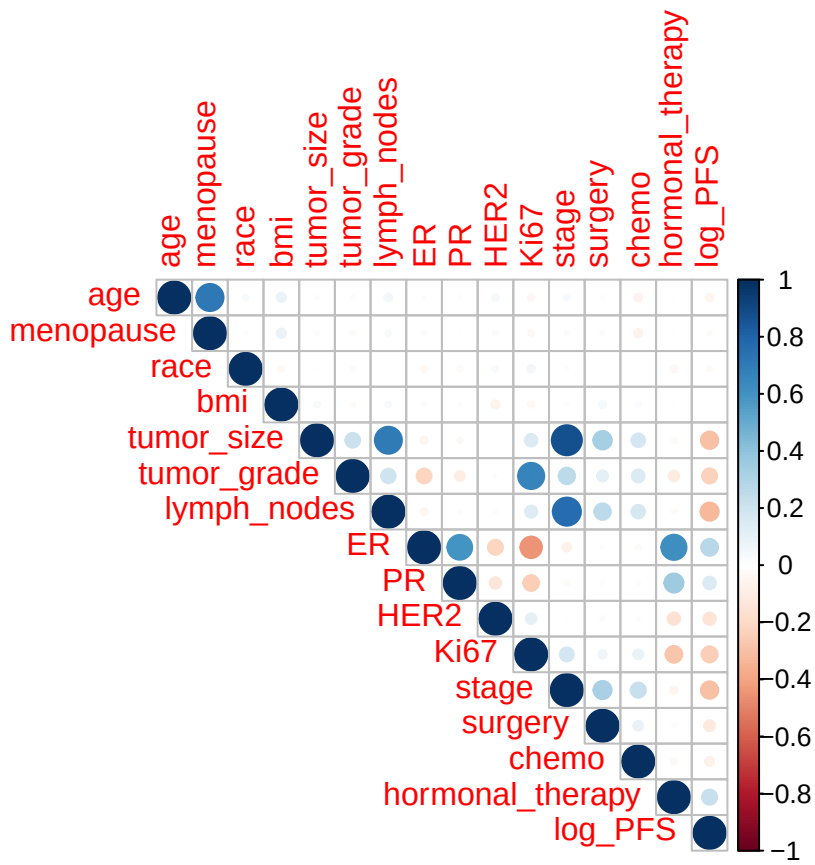
```
##       id                  age          menopause           race      
##  Length:2000        Min.   :25.00   Min.   :0.0000   Min.   :1.000  
##  Class :character   1st Qu.:47.00   1st Qu.:0.0000   1st Qu.:1.000  
##  Mode  :character   Median :55.00   Median :1.0000   Median :1.000  
##                     Mean   :54.99   Mean   :0.6555   Mean   :1.697  
##                     3rd Qu.:62.40   3rd Qu.:1.0000   3rd Qu.:2.000  
##                     Max.   :85.00   Max.   :1.0000   Max.   :4.000  
##       bmi          tumor_size     tumor_grade     lymph_nodes      
##  Min.   :16.00   Min.   :0.300   Min.   :1.000   Min.   : 0.000  
##  1st Qu.:23.00   1st Qu.:2.100   1st Qu.:2.000   1st Qu.: 0.750  
##  Median :26.05   Median :3.100   Median :2.000   Median : 2.000  
##  Mean   :26.06   Mean   :3.384   Mean   :2.182   Mean   : 2.623  
##  3rd Qu.:29.10   3rd Qu.:4.800   3rd Qu.:3.000   3rd Qu.: 4.000  
##  Max.   :42.60   Max.   :8.300   Max.   :3.000   Max.   :16.000  
##        ER               PR              HER2            Ki67      
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.000   Min.   : 1.00  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:13.50  
##  Median :1.0000   Median :0.0000   Median :0.000   Median :21.60  
##  Mean   :0.6405   Mean   :0.4805   Mean   :0.222   Mean   :21.85  
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.000   3rd Qu.:29.90  
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.000   Max.   :62.30  
##      stage         surgery          chemo        hormonal_therapy
##  Min.   :1.00   Min.   :0.000   Min.   :0.0000   Min.   :0.000  
##  1st Qu.:1.00   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.000  
##  Median :2.00   Median :1.000   Median :1.0000   Median :0.000  
##  Mean   :2.05   Mean   :0.507   Mean   :0.5525   Mean   :0.467  
```

```
##  3rd Qu.:3.00   3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.000
##  Max.   :3.00   Max.   :1.000   Max.   :1.0000   Max.   :1.000
##     log_PFS
##  Min.   :-0.5345
##  1st Qu.: 1.6724
##  Median : 2.1911
##  Mean   : 2.1957
##  3rd Qu.: 2.7125
##  Max.   : 5.1013
```

```r
# checking for any NAs in data

colSums(is.na(datA))
```

```
##               id              age        menopause             race
##                0                0                0                0
##              bmi       tumor_size      tumor_grade      lymph_nodes
##                0                0                0                0
##               ER               PR             HER2             Ki67
##                0                0                0                0
##            stage          surgery            chemo hormonal_therapy
##                0                0                0                0
##          log_PFS
##                0
```

```r
# Select only numeric columns for the matrix
numeric_data = datA |> select(where(is.numeric))

# Calculate and plot
cor_matrix = cor(numeric_data, use = "complete.obs")
corrplot(cor_matrix, method = "circle", type = "upper")
```



There are a few correlated variables, which intuitively make sense:

- `tumor_size`, `lymph_nodes`, and `stage` are highly positively correlated
- `age` and `menopause` are positively correlated
- `Ki67` (measures how fast cancer cells in a tumor are dividing) and `tumor_grade` (how abnormal cancer cells look) are positively correlated
- `Ki67` and `ER` (estrogen receptor) are negatively correlated
- `PR` (progesterone receptor) and `ER` are positively correlated
- `hormonal_therapy` and `ER` are positively correlated

```
# converting categorical variables into factors, as described in the assignment table

datA =
  datA |>
  mutate(
    menopause = factor(menopause, levels = c("0", "1")),
    race = factor(race, levels = c("1", "2", "3", "4")),
    tumor_grade = factor(tumor_grade, levels = c("1", "2", "3")),
    stage = factor(stage, levels = c("1", "2", "3")),
    surgery = factor(surgery, levels = c("0", "1")),
    chemo = factor(chemo, levels = c("0", "1")),
    hormonal_therapy = factor(hormonal_therapy, levels = c("0", "1")),
    ER = factor(ER, levels = c("0", "1")),
    PR = factor(PR, levels = c("0", "1")),
    HER2 = factor(HER2, levels = c("0", "1"))
  ) |>
```

```r
  select(-id) # removing id since it isnt predictive

# splitting data into training and testing for model training

set.seed(1)
dat_split <- initial_split(datA, prop = 0.8)
trainData <- training(dat_split)
testData <- testing(dat_split)
```

2. Fitting a linear model

Calculating VIF on training data to see if the correlations observed earlier are multicollinear:

```r
# Fit a standard linear model first
vif_model = lm(log_PFS ~ ., data = trainData)

# Calculate VIF
vif(vif_model)
```

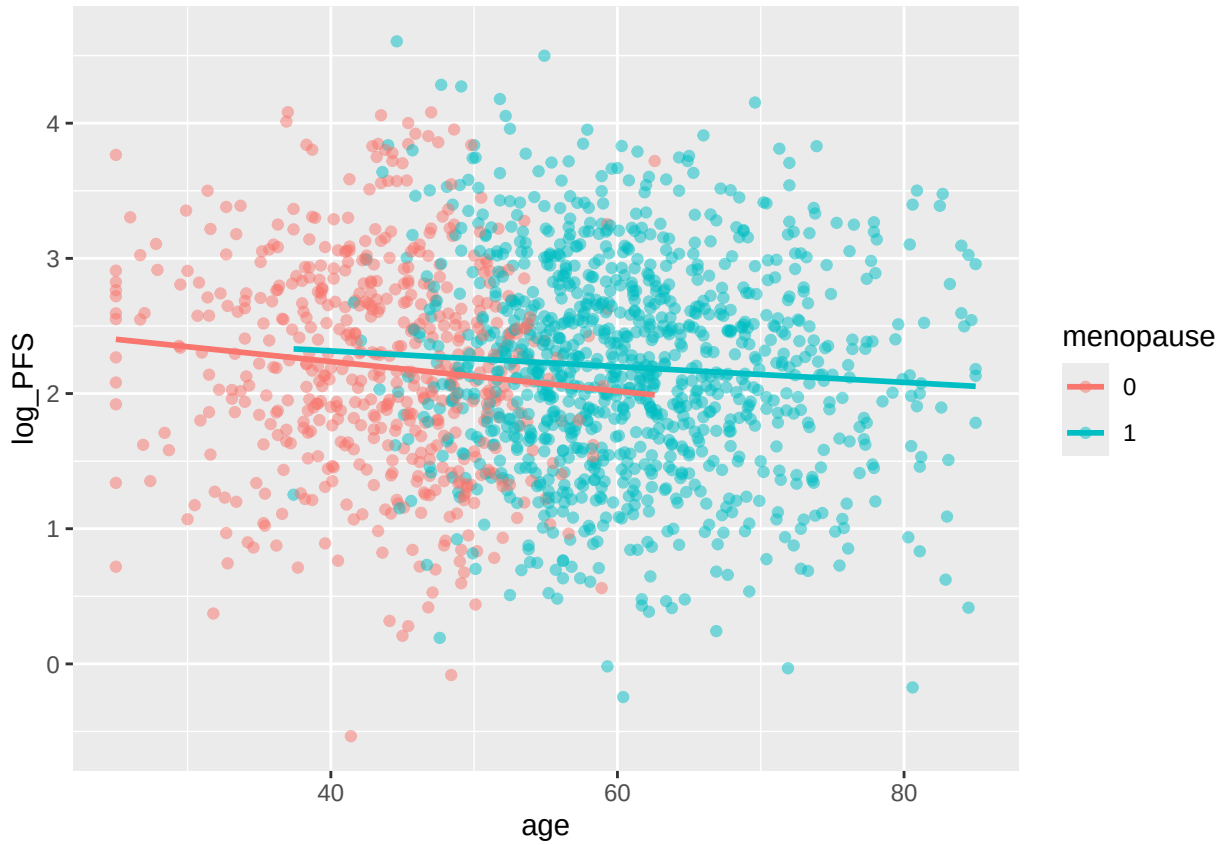
```
##                      GVIF Df GVIF^(1/(2*Df))
## age              2.120172  1        1.456081
## menopause        2.114604  1        1.454168
## race             1.030801  3        1.005069
## bmi              1.020388  1        1.010143
## tumor_size       4.712095  1        2.170736
## tumor_grade      1.988493  2        1.187493
## lymph_nodes      2.655412  1        1.629544
## ER               2.538460  1        1.593255
## PR               1.537270  1        1.239867
## HER2             1.065945  1        1.032446
## Ki67             2.192391  1        1.480673
## stage            6.315539  2        1.585268
## surgery          1.128288  1        1.062209
## chemo            1.076826  1        1.037702
## hormonal_therapy 1.634647  1        1.278533
```

No variables are multicollinear, so all can be kept in models.

Checking for interactions:

```r
ggplot(trainData, aes(x = tumor_size, y = log_PFS, color = stage)) +
    geom_point(alpha = 0.5) +
    geom_smooth(method = "lm", se = FALSE)
```

```
ggplot(trainData, aes(x = age, y = log_PFS, color = menopause)) +
    geom_point(alpha = 0.5) +
    geom_smooth(method = "lm", se = FALSE)
```



It looks like age/menopause and tumor size/cancer stage have an interaction. When we run the MARS model, we can take a closer look at these interactions.

3. Fitting lm, knn, enet

We employed 10 fold cv

```
# Using 10 fold cross validation for model comparison
set.seed(1)
folds <- createFolds(trainData$log_PFS, k = 10)
ctrl <- trainControl(method = "repeatedcv",
                     number = 10,
                     repeats = 5,
                     index = folds)

# Linear regression

set.seed(1)
fit.lm <- train(log_PFS ~ .,
                data = trainData,
                method = "lm",
                trControl = ctrl)

# K-nearest neighbors
```

```r
set.seed(1)
fit.knn <- train(log_PFS ~ .,
                 data = trainData,
                 method = "knn",
                 preProcess = c("center", "scale"),
                 tuneGrid = expand.grid(k = 1:25),
                 trControl = ctrl)

# Grabbing best params
fit.knn$bestTune
```

```
##     k
## 25 25
```

```r
# Elastic net
set.seed(1)
fit.enet <- train(log_PFS ~ .,
                  data = trainData,
                  method = "glmnet",
                  tuneGrid = expand.grid(
                    alpha = seq(0, 1, length = 11),
                    lambda = exp(seq(-5, 2, length = 50))
                  ),
                  trControl = ctrl)

# Grabbing best params
fit.enet$bestTune
```

```
##    alpha    lambda
## 34     0 0.7514773
```

4. Fitting pls, pcr, gam, mars

```r
# Prepare matrices
x_train <- model.matrix(log_PFS ~ ., trainData)[, -1]
y_train <- trainData$log_PFS
x_test <- model.matrix(log_PFS ~ ., testData)[, -1]
y_test <- testData$log_PFS

set.seed(1)
pcr_model <- train(
  log_PFS ~ .,
  data = trainData,
  method = "pcr",
  preProcess = c("center", "scale"),
  trControl = ctrl,
  tuneLength = 20
)

set.seed(1)
```

```r
pls_model <- train(
  log_PFS ~ .,
  data = trainData,
  method = "pls",
  preProcess = c("center", "scale"),
  trControl = ctrl,
  tuneLength = 20
)

# GAM
set.seed(1)
gam_model <- train(log_PFS ~ .,
                   data = trainData,
                   method = "gam",
                   trControl = ctrl)

# MARS
set.seed(1)
mars_model <- train(log_PFS ~ .,
                    data = trainData,
                    method = "earth",
                    tuneGrid = expand.grid(degree = 1:3, nprune = 2:15),
                    trControl = ctrl)
```

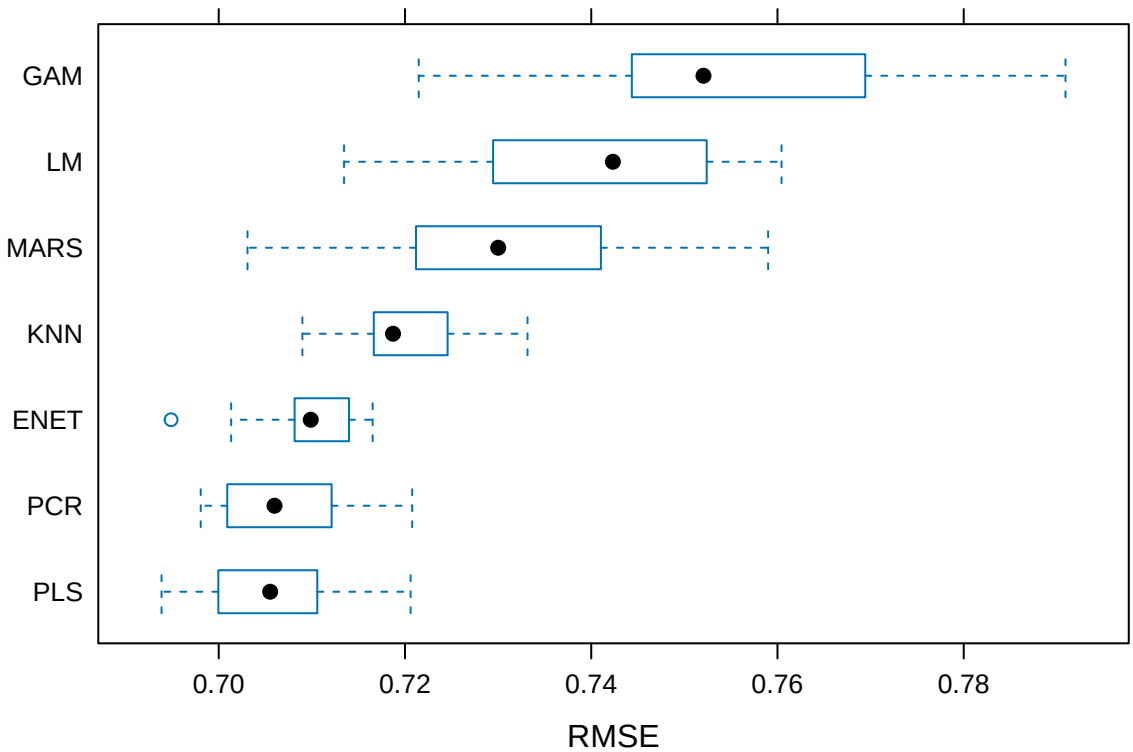
5. Comparing all models training RMSE

```r
all_models <- resamples(list(
  LM = fit.lm,
  KNN = fit.knn,
  ENET = fit.enet,
  PCR = pcr_model,
  PLS = pls_model,
  GAM = gam_model,
  MARS = mars_model
))

# Extract just the RMSE statistics matrix
rmse_matrix <- summary(all_models, metric = "RMSE")$statistics$RMSE
cv_means <- rmse_matrix[, "Mean"]
best_model_name <- names(cv_means)[which.min(cv_means)]
best_model_rmse <- min(cv_means)
worst_model_name <- names(cv_means)[which.max(cv_means)]
worst_model_rmse <- max(cv_means)

rmse_matrix |> knitr::kable(digits = 4)
```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
LM	0.7135	0.7310	0.7423	0.7402	0.7519	0.7604	0
KNN	0.7090	0.7168	0.7187	0.7200	0.7235	0.7332	0

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
ENET	0.6949	0.7083	0.7099	0.7090	0.7135	0.7165	0
PCR	0.6981	0.7010	0.7060	0.7066	0.7112	0.7208	0
PLS	0.6939	0.7000	0.7055	0.7054	0.7099	0.7206	0
GAM	0.7215	0.7457	0.7521	0.7542	0.7661	0.7909	0
MARS	0.7031	0.7214	0.7300	0.7305	0.7407	0.7590	0

```r
bwplot(all_models, metric = "RMSE")
```




```r
cv_rmse = summary(all_models)$statistics$RMSE[, "Mean"]

# evaluating test predictions

predictions <- list(
  LM = predict(fit.lm, testData),
  KNN = predict(fit.knn, testData),
  ENET = predict(fit.enet, testData),
  PCR = predict(pcr_model, testData),
  PLS = predict(pls_model, testData),
  MARS = predict(mars_model, testData),
  GAM = predict(gam_model, testData)
)

# Post-resample RMSE on test set
```

```r
test_rmse <- sapply(predictions, function(pred) postResample(pred, testData$log_PFS)[["RMSE"]])

# summarize RMSE in table
model_comparison = tibble(
  model = names(test_rmse),
  cv_rmse = cv_rmse,
  test_rmse = test_rmse
  ) |>
  arrange(test_rmse)
model_comparison |> knitr::kable(digits = 3)
```

model	cv_rmse	test_rmse
GAM	0.730	0.741
ENET	0.709	0.744
PLS	0.705	0.745
KNN	0.720	0.746
PCR	0.707	0.748
LM	0.740	0.749
MARS	0.754	0.751

Final Coefficients:

```r
enet_coefs = coef(fit.enet$finalModel, fit.enet$bestTune$lambda) |> as.matrix()

enet_coefs_clean = tibble(
  predictor = rownames(enet_coefs),
  coefficient = enet_coefs[, 1]
  ) |>
  arrange(desc(abs(coefficient)))

enet_coefs_clean |> knitr::kable(digits = 3)
```

predictor	coefficient
(Intercept)	2.550
ER1	0.129
hormonal_therapy1	0.106
stage3	-0.102
HER21	-0.101
tumor_grade3	-0.088
PR1	0.038
tumor_size	-0.034
surgery1	-0.033
race4	-0.032
lymph_nodes	-0.030
race3	0.028
race2	0.019
menopause1	0.015
chemo1	-0.012
tumor_grade2	-0.006

predictor	coefficient
Ki67	-0.004
age	-0.002
bmi	0.000
stage2	0.000

Warning: Some resampling iterations for the elastic net model produced missing performance values, likely due to numerical instability at extreme tuning parameter values. However, the majority of resamples were valid, and the selected tuning parameters were stable.

We used repeated 10-fold cross-validation to estimate out-of-sample prediction error. This approach repeatedly partitions the training data into training and validation sets, reducing variance compared to a single split. We evaluated seven candidate models: Linear Regression, K-Nearest Neighbors, Elastic Net (ENET), Principal Component Regression (PCR), Partial Least Squares (PLS), Generalized Additive Models (GAM), and Multivariate Adaptive Regression Splines (MARS).

Based on the training data, most models performed very similarly, with mean RMSE values ranging from 0.7054 (PLS) to 0.7542 (GAM). The similar performance across linear and nonlinear models suggests that the relationship between the predictors and `log(PFS)` is largely linear.

PLS achieved the lowest mean error, closely followed by PCR (Mean = 0.7066) and ENET (Mean = 0.709). ENET and PCR performing similarly to PLS suggests that dimensionality reduction provided little additional predictive power. ENET has the lowest maximum RMSE observed. With the exception of one outlier, the remaining RMSEs from cross-validation have a tight distribution, indicating this model is less variable and more stable.

Because predictive accuracy was comparable across the top models, we prioritized interpretability. We selected the ENET model, which was trained using an explicit lambda grid to ensure consistent tuning across folds. The cross-validation process selected an alpha of 0, meaning the final model is a pure Ridge regression. While Ridge regression does not perform sparse feature selection (all variables are retained), it stabilizes the coefficients to handle multicollinearity, making it a highly robust and easily interpretable model for our clinical context. The reported test error for this ENET model was 0.7445.

6. Load, clean, and prepare dataset B

```r
load("data/datB.RData")
summary(datB)
```

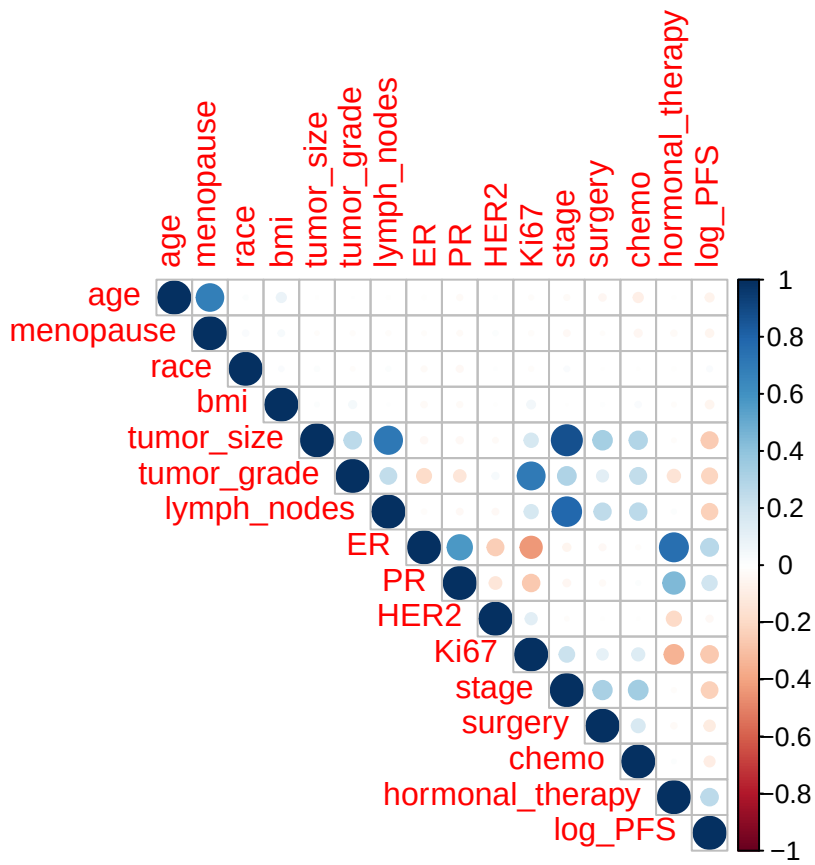
```
##       id                 age          menopause          race
##  Length:2000        Min.   :25.00   Min.   :0.000   Min.   :1.000
##  Class :character   1st Qu.:48.50   1st Qu.:0.000   1st Qu.:1.000
##  Mode  :character   Median :56.00   Median :1.000   Median :1.000
##                     Mean   :56.05   Mean   :0.681   Mean   :1.875
##                     3rd Qu.:63.80   3rd Qu.:1.000   3rd Qu.:3.000
##                     Max.   :85.00   Max.   :1.000   Max.   :4.000
##       bmi          tumor_size     tumor_grade     lymph_nodes
##  Min.   :16.00   Min.   :0.300   Min.   :1.000   Min.   : 0.000
##  1st Qu.:24.00   1st Qu.:1.800   1st Qu.:1.000   1st Qu.: 0.000
##  Median :27.30   Median :2.800   Median :2.000   Median : 2.000
##  Mean   :27.44   Mean   :3.075   Mean   :2.079   Mean   : 2.318
##  3rd Qu.:30.82   3rd Qu.:4.200   3rd Qu.:3.000   3rd Qu.: 4.000
```

```
##  Max.   :45.90   Max.   :7.900   Max.   :3.000   Max.   :15.000
##        ER              PR              HER2            Ki67
##  Min.   :0.000   Min.   :0.0000   Min.   :0.000   Min.   : 1.00
##  1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:11.70
##  Median :1.000   Median :1.0000   Median :0.000   Median :19.85
##  Mean   :0.674   Mean   :0.5045   Mean   :0.237   Mean   :20.37
##  3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:0.000   3rd Qu.:28.80
##  Max.   :1.000   Max.   :1.0000   Max.   :1.000   Max.   :62.30
##      stage          surgery           chemo       hormonal_therapy
##  Min.   :1.000   Min.   :0.0000   Min.   :0.000   Min.   :0.0000
##  1st Qu.:1.000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:0.0000
##  Median :2.000   Median :0.0000   Median :1.000   Median :1.0000
##  Mean   :1.894   Mean   :0.4415   Mean   :0.649   Mean   :0.5695
##  3rd Qu.:2.000   3rd Qu.:1.0000   3rd Qu.:1.000   3rd Qu.:1.0000
##  Max.   :3.000   Max.   :1.0000   Max.   :1.000   Max.   :1.0000
##     log_PFS
##  Min.   :-0.0866
##  1st Qu.: 2.2536
##  Median : 2.7440
##  Mean   : 2.7424
##  3rd Qu.: 3.2487
##  Max.   : 5.2864
```

```r
# checking for any NAs in data

colSums(is.na(datB))
```

```
##               id              age        menopause             race
##                0                0                0                0
##              bmi       tumor_size      tumor_grade      lymph_nodes
##                0                0                0                0
##               ER               PR             HER2             Ki67
##                0                0                0                0
##            stage          surgery            chemo hormonal_therapy
##                0                0                0                0
##          log_PFS
##                0
```

```r
# Select only numeric columns for the matrix
numeric_data_B = datB |> select(where(is.numeric))

# Calculate and plot
cor_matrix_B = cor(numeric_data_B, use = "complete.obs")
corrplot(cor_matrix_B, method = "circle", type = "upper")
```



Correlations are similar to datA. Proceeding with same data cleaning

```
# converting categorical variables into factors, as described in the assignment table

datB =
  datB |>
  mutate(
    menopause = factor(menopause, levels = c("0", "1")),
    race = factor(race, levels = c("1", "2", "3", "4")),
    tumor_grade = factor(tumor_grade, levels = c("1", "2", "3")),
    stage = factor(stage, levels = c("1", "2", "3")),
    surgery = factor(surgery, levels = c("0", "1")),
    chemo = factor(chemo, levels = c("0", "1")),
    hormonal_therapy = factor(hormonal_therapy, levels = c("0", "1")),
    ER = factor(ER, levels = c("0", "1")),
    PR = factor(PR, levels = c("0", "1")),
    HER2 = factor(HER2, levels = c("0", "1"))
  ) |>
  select(-id) # removing id since it isnt predictive
```

7. Testing ENET model on dataset B

```
enet_pred_B = predict(fit.enet, newdata = datB)
enet_test_rmse <- RMSE(pred = enet_pred_B, obs = datB$log_PFS)
```

```r
print(enet_test_rmse)
```

```
## [1] 0.8422413
```

When applying the final Elastic Net model (trained on datA) to the new cohort (datB), the RMSE increased to 0.8422. Compared to the previous test RMSE of 0.7445 on the original data, this indicates a drop in predictive performance. This gap suggests that there may be underlying distribution shifts between the two patient populations (e.g., differences in demographics, clinical settings, or unmeasured variables), or that the model slightly overfit the original cohort despite the use of Ridge regularization. A new model will be needed. Since ENET had a tight distribution when using the original data, it seems the most promising (rather than training many different models). Additionally, it is then easier to compare what changed in the model.

8. Trainning new model for dataset B

```r
# splitting data into training and testing for model training

set.seed(1)
dat_split <- initial_split(datB, prop = 0.8)
trainData <- training(dat_split)
testData <- testing(dat_split)

# Elastic net
set.seed(1)
fit.enet.new <- train(log_PFS ~ .,
                      data = trainData,
                      method = "glmnet",
                      tuneGrid = expand.grid(
                        alpha = seq(0, 1, length = 11),
                        lambda = exp(seq(-5, 5, length = 50))
                        ),
                      trControl = ctrl)

# Grabbing best params
fit.enet.new$bestTune
```

```
##    alpha    lambda
## 24     0 0.7362965
```

```r
enet_new_pred = predict(fit.enet.new, newdata = testData)
enet_new_test_rmse <- RMSE(pred = enet_new_pred, obs = testData$log_PFS)
print(enet_new_test_rmse)
```

```
## [1] 0.7104038
```

This RMSE looks more similar to the test RMSE from the original data.

Comparing the old and new model:

```r
enet_old_coefs = as.matrix(coef(fit.enet$finalModel, fit.enet$bestTune$lambda))

coefs_old = tibble(
  predictor = rownames(enet_old_coefs),
  coefficient = enet_old_coefs[, 1]) |>
  arrange(desc(abs(coefficient)))

coefs_old |> knitr::kable(digits = 4)
```

predictor	coefficient
(Intercept)	2.5499
ER1	0.1286
hormonal_therapy1	0.1058
stage3	-0.1017
HER21	-0.1009
tumor_grade3	-0.0881
PR1	0.0384
tumor_size	-0.0343
surgery1	-0.0325
race4	-0.0318
lymph_nodes	-0.0300
race3	0.0284
race2	0.0195
menopause1	0.0149
chemo1	-0.0119
tumor_grade2	-0.0065
Ki67	-0.0042
age	-0.0021
bmi	-0.0002
stage2	0.0000

```r
enet_new_coefs = as.matrix(coef(fit.enet.new$finalModel, fit.enet.new$bestTune$lambda))

coefs_new = tibble(
  predictor = rownames(enet_new_coefs),
  coefficient = enet_new_coefs[, 1]) |>
  arrange(desc(abs(coefficient)))

coefs_new |> knitr::kable(digits = 4)
```

predictor	coefficient
(Intercept)	3.0341
hormonal_therapy1	0.1224
ER1	0.1117
stage3	-0.0842
tumor_grade3	-0.0822
PR1	0.0611
tumor_grade2	0.0568
race4	0.0478
surgery1	-0.0476

predictor	coefficient
tumor_size	-0.0293
race3	-0.0284
menopause1	-0.0220
lymph_nodes	-0.0158
chemo1	-0.0147
race2	-0.0119
stage2	-0.0105
HER21	0.0086
Ki67	-0.0041
bmi	-0.0031
age	-0.0019

```
merged_coefs =
  full_join(coefs_old, coefs_new,
            by = "predictor", suffix = c("_old", "_new")) |>
  mutate(difference = coefficient_old - coefficient_new)

merged_coefs |> arrange(predictor)  |> knitr::kable(digits = 4)
```

predictor	coefficient_old	coefficient_new	difference
(Intercept)	2.5499	3.0341	-0.4842
ER1	0.1286	0.1117	0.0169
HER21	-0.1009	0.0086	-0.1096
Ki67	-0.0042	-0.0041	-0.0001
PR1	0.0384	0.0611	-0.0227
age	-0.0021	-0.0019	-0.0002
bmi	-0.0002	-0.0031	0.0028
chemo1	-0.0119	-0.0147	0.0027
hormonal_therapy1	0.1058	0.1224	-0.0166
lymph_nodes	-0.0300	-0.0158	-0.0142
menopause1	0.0149	-0.0220	0.0369
race2	0.0195	-0.0119	0.0314
race3	0.0284	-0.0284	0.0567
race4	-0.0318	0.0478	-0.0796
stage2	0.0000	-0.0105	0.0105
stage3	-0.1017	-0.0842	-0.0176
surgery1	-0.0325	-0.0476	0.0150
tumor_grade2	-0.0065	0.0568	-0.0633
tumor_grade3	-0.0881	-0.0822	-0.0059
tumor_size	-0.0343	-0.0293	-0.0050

By comparing the coefficients of the old pre-2000 model to the new 2000-2005 model, several major predictive shifts become clear. The new model's intercept increased from 2.55 to 3.03, indicating that the baseline PFS increased over the decade. This makes sense since new and better interventions will have likely been more accessible. The impact of HER2 status (`HER21`) has flipped. In the older cohort, being HER2+ was a severe negative predictor for progression-free survival (-0.101). In the new cohort, this penalty essentially vanished (+0.009). Other predictors that flipped were all race categories, `menopause1`, `stage2`, and `tumor_grade2`. Others like ER and hormonal therapy stayed largely the same and continued to improve PFS, while stage 3, tumor grade 3, and tumor size decreased PFS. The gap in predictive power makes sense given these changes in cohort trends.